

Benjamin Levine

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EDUCATION

Stony Brook University

Doctorate in Physics (Concentration in Astronomy)

Stony Brook, NY

Expected May 2027

The University of Chicago

Bachelor of Science in Astrophysics (Honors)

Chicago, IL

June 2022

PUBLICATIONS & MANUSCRIPTS

First Author & Major Contributions

- “Galaxy Clustering with LSST: Effects of Number Count Bias from Blending,” **B. Levine**, *et al.*, OJA 8, 2025.
- “Brightest Cluster Galaxies in Strongly-Lensing Galaxy Clusters and Richness- and Redshift-Matched Samples,” **B. Levine**, Bachelor’s Thesis, The University of Chicago, 2022.

Contributing Author

- “Constraining Galaxy Cluster Triaxiality via Weak Lensing – I. Preparation for the Rubin Data Beyond Leading Order,” S. Fu, ..., **B. Levine**, *et al.*, Submitted to ApJ, 2026.
- “Brightest Cluster Galaxy ellipticity as proxy for halo shape: Orientation bias, assembly bias, and potential selection effects in SZ-selected clusters,” R. Srinivasan, ..., **B. Levine**, *et al.*, Submitted to OJA, 2026.
- “Halo Occupation Distribution estimation performance for LSST data,” P. Cataldi, ..., **B. Levine**, *et al.*, A&A 712 A114, 2026.
- “The Vera C. Rubin Observatory Data Preview 1,” Vera C. Rubin Observatory Team, ..., **B. Levine**, *et al.*, AJ 171 360, 2026.
- “NSF-DOE Vera C. Rubin Observatory Observations of Interstellar Comet 3I/ATLAS (C/2025 N1),” C. Chandler, ..., **B. Levine**, *et al.*, ApJL 1001 L35, 2026.
- “COOL-LAMPS. VIII. Known wide-separation lensed quasars and their host galaxies reveal a lack of evolution in $M_{BH}/M_{\star}$ since $z \sim 3$,” A. Cloonan, ..., **B. Levine**, *et al.*, ApJ 987 194, 2025.
- “COOL-LAMPS. VII. Quantifying Strong-lens Scaling Relations with 177 Cluster-scale Gravitational Lenses in DECaLS,” S. Mork, ..., **B. Levine**, *et al.*, ApJ 979 184, 2025.
- “Rubin M1M3 support system dynamic performance,” B. Quint, ..., **B. Levine**, *et al.*, Proc. SPIE 13094, Ground-based and Airborne Telescopes X, 1309429, 2024.
- “The impact of human expert visual inspection on the discovery of strong gravitational lenses,” K. Rojas, ..., **B. Levine**, *et al.*, MNRAS 523 3, 2023.
- “COOL-LAMPS. IV. A Sample of Bright Strongly-Lensed Galaxies at $3 < z < 4$,” Y. Zhang, Y., ..., **B. Levine**, *et al.*, ApJ 950 58, 2023.

AWARDS & HONORS

Stony Brook University

Gerald Brown Prize for Outstanding Research

2024, 2026

Excellence in Teaching - Teaching Assistant Award

2024

David Fox Prize for Outstanding Teaching Assistant

2023

Lourie Summer Research Fellowship

2023

U.S. Department of Energy

Office of Science Graduate Student Research (SCGSR) Award 2026

State University of New York System

SUNY Graduate Research Empowering and Accelerating Talent (GREAT) Award 2025

National Science Foundation

NSF Graduate Research Fellowship Program (GRFP) Honorable Mention 2024

The University of Chicago

Dean's List 2019, 2022

Lawrence A. Kimpton Fellowship 2022

Summer Research Scholar Grant 2021

National Merit Scholarship Program

National Merit Scholarship 2018–2022

TEACHING EXPERIENCE

Stony Brook University

Lead Instructor: High School Women in Science and Engineering (HS WISE) Fall 2023–Spring 2026

Guest Lecturer: HS WISE Computational Astrophysics Course Spring 2025

Guest Lecturer: AST200 — Current Astronomical Research at Stony Brook Spring 2025

Teaching Assistant: AST443/PHY517 — Observational Techniques in Astronomy Fall 2022–Spring 2024

The University of Chicago

Teaching Assistant: ASTR119 — Physics of Stars: An Introduction Summer 2021

TALKS

Plenary Science Talk: “Modeling and Forecasting for DP2 Galaxy-Galaxy Lensing”

LSST Dark Energy Science Collaboration, Summer Meeting July 2026

“Modeling Progress for the LSST DESC Galaxy-Galaxy Lensing Key Project”

LSST Dark Energy Science Collaboration, Summer Meeting July 2026

“How to Measure the Structure of the Universe”

Stony Brook University, AST200 Course February 2025

“Galaxy Clustering with LSST: Effects of Number Count Bias from Blending”

LSST Dark Energy Science Collaboration November 2024

“The Effects of Blending on LSST Galaxy Clustering Analyses”

LSST Dark Energy Science Collaboration, Summer Meeting July 2023

LSST Dark Energy Science Collaboration, Blending Working Group July 2023

“Brightest Cluster Galaxies in Strongly-Lensing Galaxy Clusters and Richness- and Redshift-Matched Samples”

The University of Chicago, Department of Astronomy & Astrophysics May 2022

“Studying the Effects of Blending in Galaxy Clustering”

LSST Dark Energy Science Collaboration, Blending Working Group September 2020

The University of Chicago, Survey Science Group September 2020

POSTER PRESENTATIONS

“Searching for Halo Ellipticity Bias in Galaxy Survey Cosmology”

Stony Brook University BRIDGES Conference February 2026

“The Effects of Blending on LSST Galaxy Clustering Analyses”

LSST Dark Energy Science Collaboration Summer Meeting

July 2023

“COOL-LAMPS: Investigating Bright Central Galaxy Luminosities in Strongly Lensing Clusters between $0.2 < z < 1.5$ ”

239th American Astronomical Society Winter Meeting (*canceled due to COVID*)

Invited January 2022

OPEN SOURCE SCIENTIFIC SOFTWARE

DECaLS Cluster Analyzer

Customizable, multithreaded pipeline to estimate the redshift and richness of galaxy clusters at arbitrary positions in the DECaLS survey.

DECaLS Catalog Downloader

Script to quickly download large batches of catalogs from the DECaLS survey data server; also compatible with use in Python scripts.

DSF

Python package to model the galaxy-galaxy lensing $\Delta\Sigma$ datavector and its associated covariance, as well as perform forecasts for arbitrary survey properties. Used to make analysis choices for a LSST Data Preview 2 Key Project.

Red Sequence Photometric Redshift Estimator

Fast, vectorized algorithm that estimates the photometric redshift of red sequence galaxies based on a predefined color-magnitude-redshift calibration.

Scamper

Fully-automated pipeline to match image WCS headers and update catalog coordinates given a reference image or catalog, using the Astromatic Scamp software.

LEADERSHIP, SERVICE & OUTREACH

pyOpenSci Organization

Peer Reviewer (1 software package)

September 2026–Present

Stony Brook University Department of Physics and Astronomy

Physics Graduate Student Association Quality of Life Survey Committee Member

October 2025–Present

Graduate Student Peer Mentor

August 2025–August 2026

LSST Dark Energy Science Collaboration

Speakers Bureau Committee Member

February 2025–Present

Internal Publication Referee (2 papers)

October 2025–Present

Stony Brook University High School Women in Science and Engineering

Lead Instructor

Fall 2023–Spring 2026

Custer Institute and Observatory

Volunteer

August 2023–April 2026

Stony Brook University 2025 Excellence in Teaching Awards Committee

Award Selection Committee Member

April 2025